

DOC-Lingo DBMS Test Document

A controlled test document for multilingual and cross-lingual RAG evaluation.

Page 1 — DBMS Fundamentals

Database Management System (DBMS)

A Database Management System is software used to create, store, organize, retrieve, and manage data in databases. A DBMS provides controlled access to data and supports insertion, updating, deletion, and querying.

Core DBMS Functions

Common responsibilities include data storage, retrieval, transaction management, concurrency control, backup and recovery, authorization, and integrity enforcement. A DBMS separates application programs from physical storage details.

Relational Databases

A relational database represents information using tables made of rows and columns. A row represents a record, while a column represents an attribute. Keys are used to identify records and establish relationships between tables.

Important Keys

Primary key: uniquely identifies each row. **Foreign key:** references a key in another table and establishes a relationship. **Candidate key:** a unique attribute or combination of attributes eligible to become a primary key.

Example

A Student table may contain StudentID, Name, Department, and Email. StudentID can be the primary key. If an Enrollment table contains StudentID, that column can be a foreign key referencing Student.StudentID.

Page 2 — SQL, Normalization, and Transactions

SQL

Structured Query Language (SQL) is commonly used with relational databases. SELECT retrieves data, INSERT adds rows, UPDATE modifies rows, and DELETE removes rows. CREATE, ALTER, and DROP are commonly used for schema definition.

Normalization

Normalization organizes relational data to reduce unnecessary duplication and update anomalies. First Normal Form (1NF) requires atomic values. Second Normal Form (2NF) requires 1NF and removal of partial dependency on a composite key. Third Normal Form (3NF) requires 2NF and removal of transitive dependencies among non-key attributes.

Transactions and ACID

A transaction is a logical unit of database work. ACID means Atomicity, Consistency, Isolation, and Durability. Atomicity treats a transaction as a unit; Consistency preserves database rules; Isolation limits interference between concurrent transactions; Durability ensures committed changes survive failures.

Indexes

An index is an auxiliary data structure that can speed up suitable data retrieval. Indexes can reduce search work but require additional storage and may increase the cost of inserts, updates, and deletes.

Concurrency Control

Concurrency control coordinates simultaneous transactions so database consistency is preserved. Locking is one common approach. Shared locks can support reading and exclusive locks can support writing, depending on the database system and isolation requirements.

Page 3 — Security, Recovery, and Test Questions

Database Security

Database security includes authentication, authorization, access control, encryption, auditing, and protection against unauthorized modification or disclosure. Least privilege limits users and applications to the permissions required for their tasks.

Backup and Recovery

Backup creates a recoverable copy of database information. Recovery procedures restore the database after hardware problems, software errors, accidental deletion, or other disruptions. Recovery mechanisms commonly work with transaction logs and checkpoints.

DBMS Test Questions

1. What is a DBMS and what are its main functions?
2. What is the difference between a primary key and a foreign key?
3. Explain 1NF, 2NF, and 3NF in simple terms.
4. What does ACID mean in database transactions?
5. How can an index improve query performance, and what trade-off does it introduce?
6. Explain database security using authentication, authorization, and least privilege.
7. What is concurrency control and why is it needed?
8. Hindi test: DBMS kya hai aur iska mukhya upyog kya hai?
9. Hindi test: Primary key aur foreign key mein kya antar hai?
10. Hinglish test: Normalization database me duplicate data ko kaise reduce karta hai?
11. Hinglish test: ACID properties transaction ko reliable kaise banati hain?
12. Cross-lingual test: Ask in Hindi about indexes and answer using the English explanation on Page 2.
13. No-context test: What is the capital of France? This information is intentionally absent from the document.